



Coversheet

MODEL NO. : TL127VVMS01-00

ISSUED DATE: 2022-8-24

VERSION : Ver 01

- ☒ Preliminary Specification
☐ Final Product Specification

Customer : CL1

Approved by	Notes

XIAMEN TIANMA Confirmed :

Prepared by	Checked by	Approved by

This technical specification is subjected to change without notice

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Record of Revision

Rev	Issued Date	Description	Editor
01	2022-08-24	Preliminary Specification	Wang peng
02	2022-10-12	Update dimension & FOG PIN assignment & electrical specification & optical characteristics & packing method	Wang peng



1 General Specifications

1-1 panel

Item	Specification	Unit
Substrate Thickness	0.2	mm
Panel Outline Dimension	276.7154 (H)×177.3096 (V)	mm
Active Screen Size(Diagonal)	12.7	inch
Active Area	273.6154(H)* 171.0096(V)	mm
Resolution	2944(RGB) x1840	pixel
Pixel Arrangement	RGB	-
Pixel Driving Element	LTPS SFT	-
Left board	1.55	mm
Right board	1.55	mm
Up board	1.55	mm
Down board	4.75	mm
Cell Gap	3.0	um
Viewing Direction	All	O'clock
Gray Scale Inversion Direction	All	O'clock
Suggested Driver IC	NT36532	

Note 1: Viewing direction for best image quality is different from TFT definition, there is a 180 degree shift.

Note 2 : Requirements on Environmental Protection: Q/S0002

Note 3 : LCD weight tolerance : +/- 5%



Item		Specification	Unit
A/A Center~LCM outline	L	139.008(with cell tape)	mm
	R	139.008(with cell tape)	mm
	U	87.705 (with cell tape)	mm
	D	91.355 (with cell tape)	mm
LCM border (AA outline~LCM outline)	L	2.2	mm
	R	2.2	mm
	U	2.2	mm
	D	5.85	mm
Horizontal dimension		278.02(with cell tape)	mm
Vertical dimension		179.06 (with cell tape)	mm
FPC Bending/Flat width		1.13max (with cell tape)	mm
Thickness(typ)		1.85±0.15(With cell tape)	mm
Outline dimension tolerance		±0.2	mm
Cover Lens View Area Size		274.2156*171.6096	mm
Cover Lens Outline Size		TBD	mm
TP Total Thickness		0.55	mm
OCA		0.2t	mm
Cover Lens Surface Treatment		AF Coating	
Surface Hardness		7H	
Lens CS		>=700	Mpa
Lens DOL		>=35	Um
CT		<=82	Mpa
4 Point Bending		>600	Mpa
Sensor Substrate		Sensor in TFT Glass	
Sensor pitch		H 4740/4554* V 4275	um
Number of touch		10	
Glove touch		TBD	
Channels		TX60, RX40	橫屏
Accuracy		6phi: Centre=1.0, Border=1.5	mm

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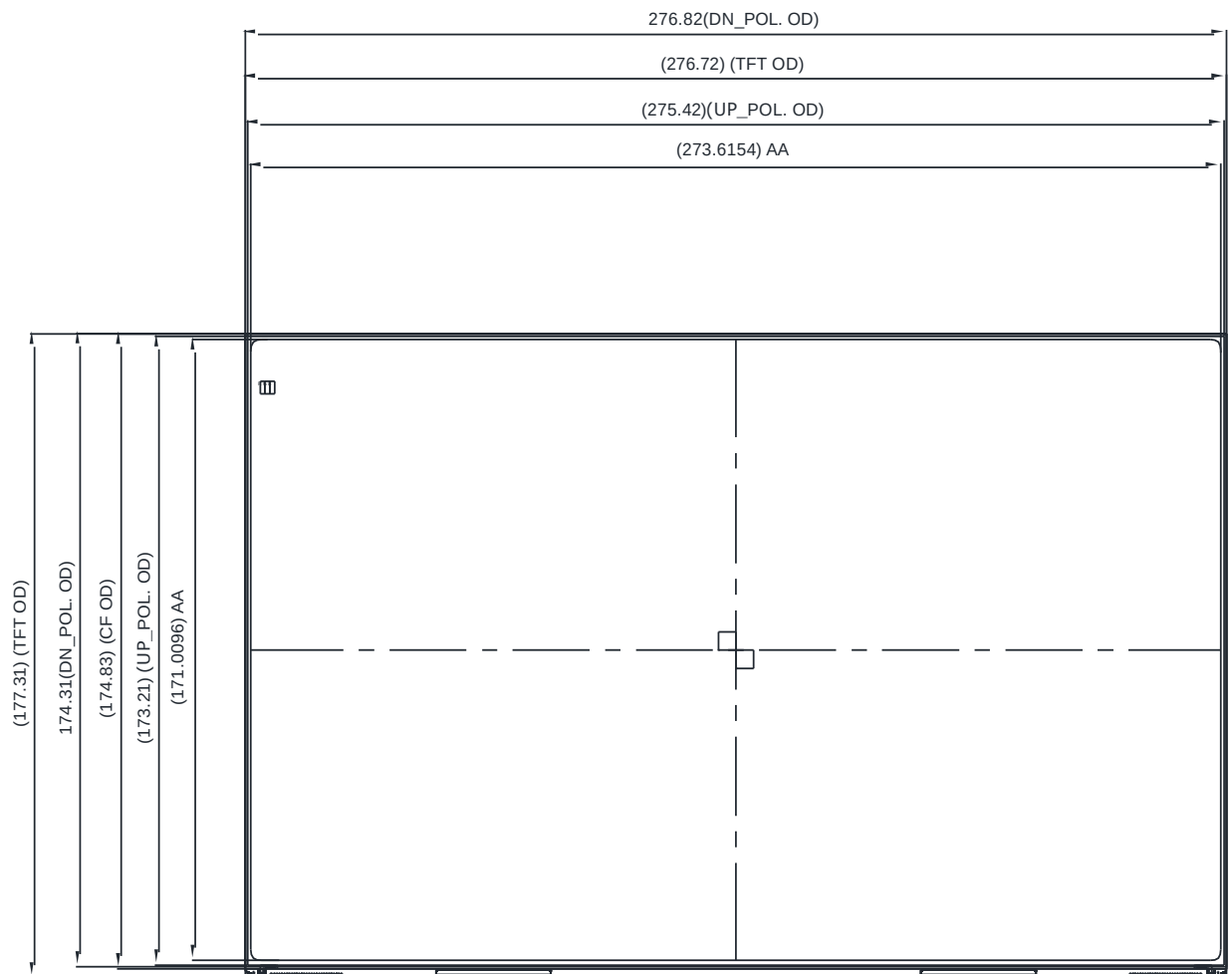


Precision	6phi:Centre=1.0, Border=1.5	mm
Linearity	6phi:Centre=1.0, Border=1.5	mm
Finger Separation	<15(center to center)	mm
Report Rate	120Hz@Only finger	Hz
Latency	20 Average/ 25 Max (depends on panel)	ms
Signal-to-Noise Ratio	40db	

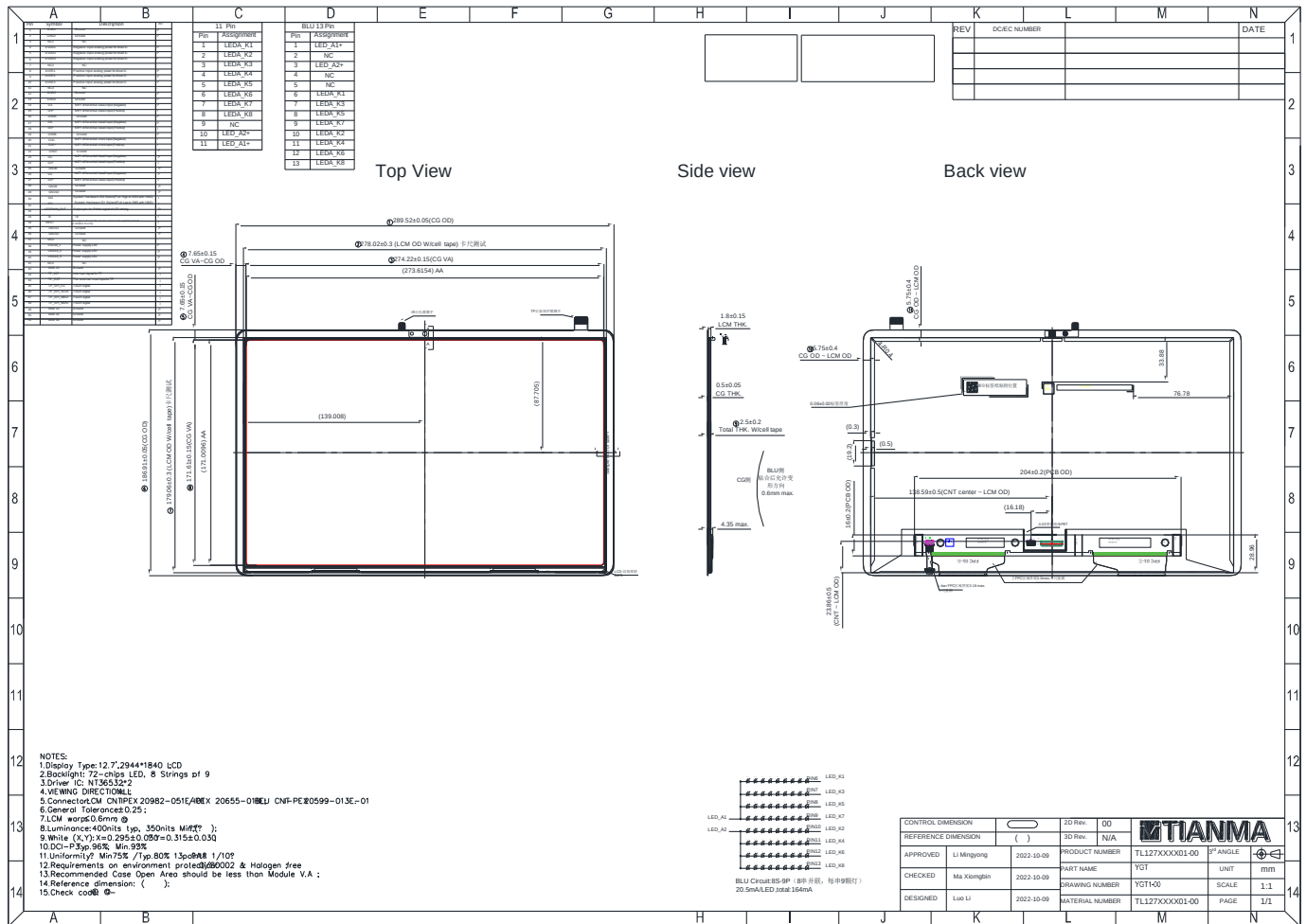


2 Dimension

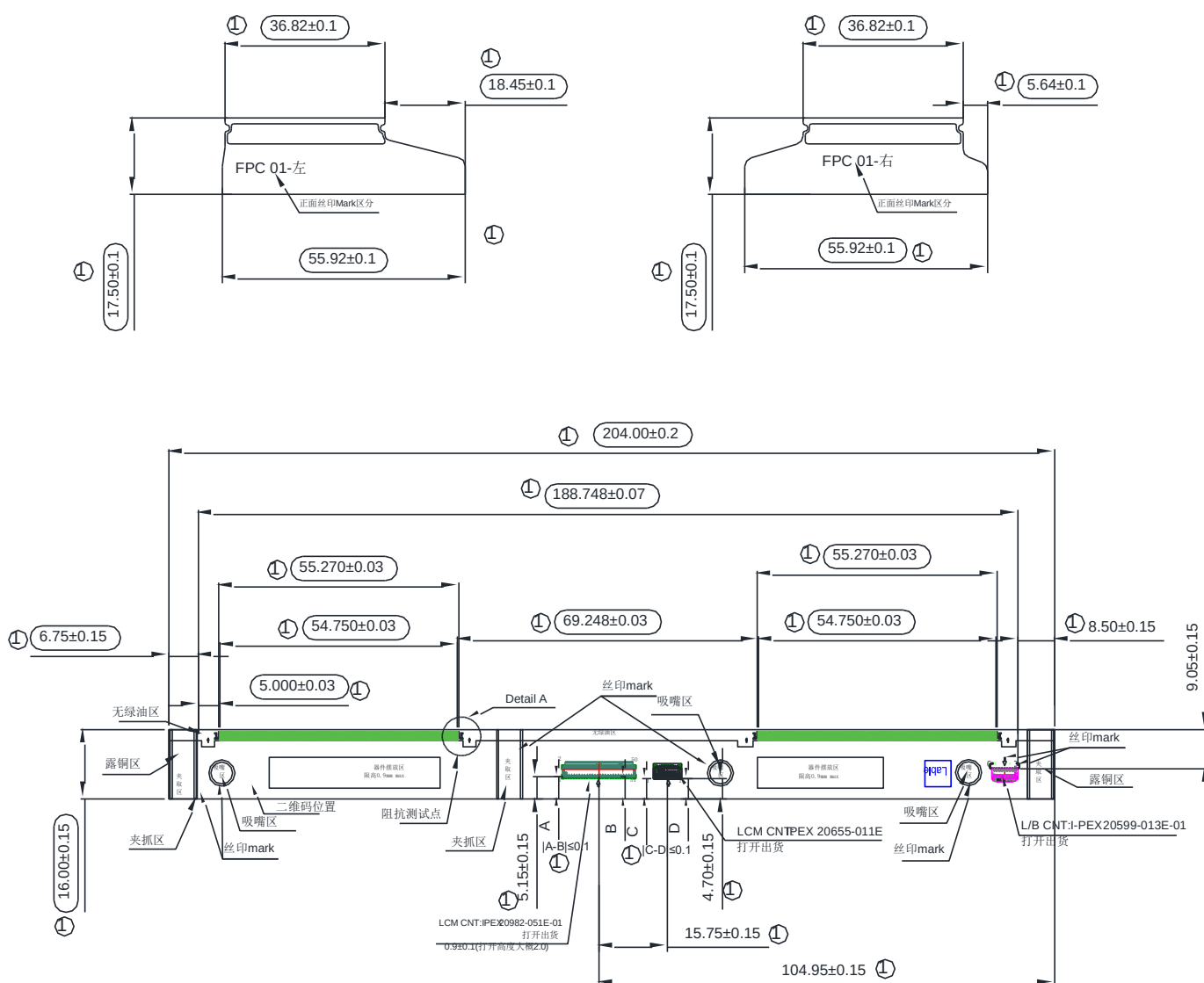
2.1 Cell Outline Dimension



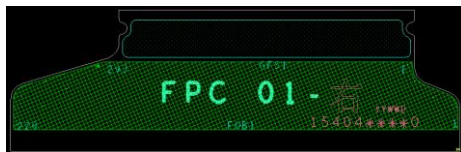
LCM Dimension



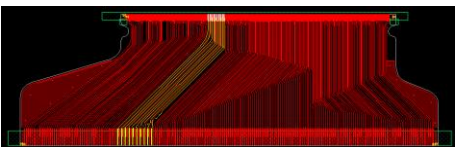
2.2 FPC Detail



TOP

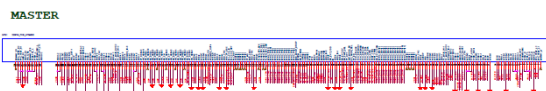


BOT

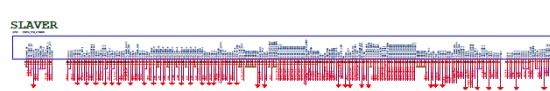


2.3 Test Pad

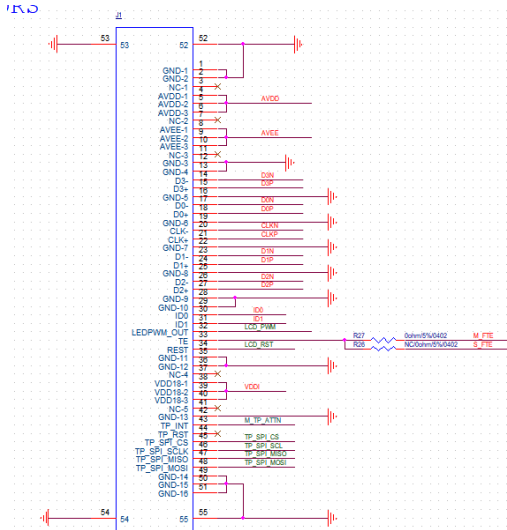
FOB1



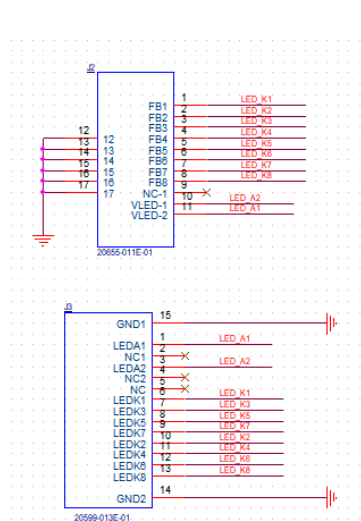
FOB2



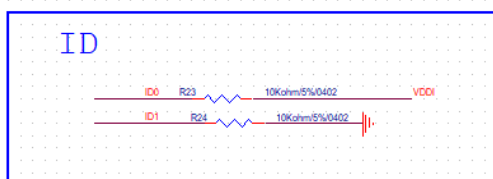
Main CNT



LED CNT



ID

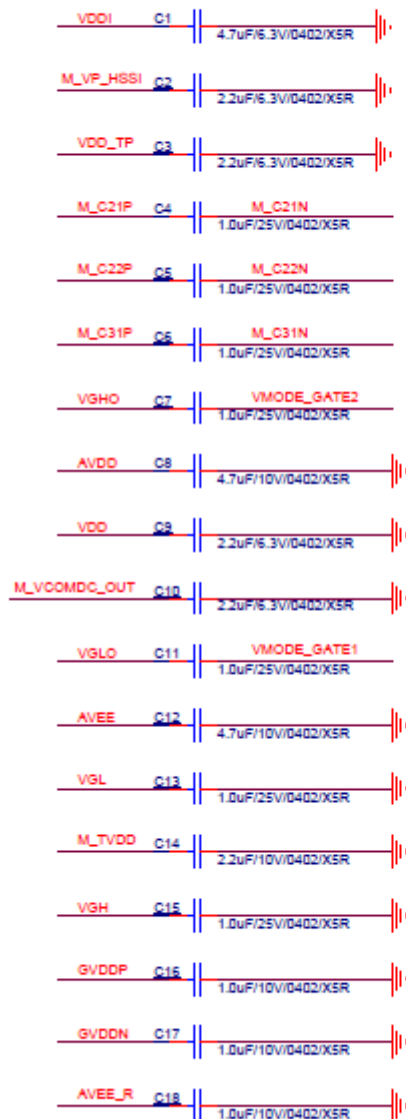


Capacitor

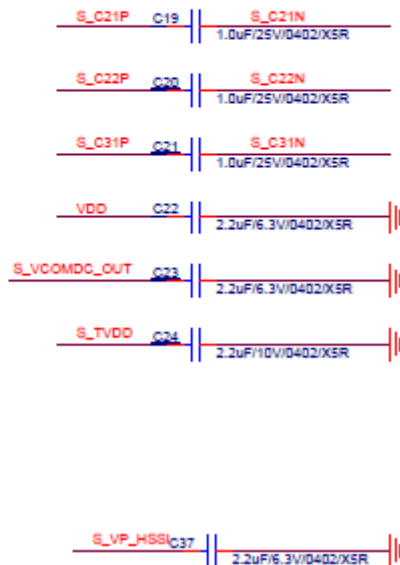


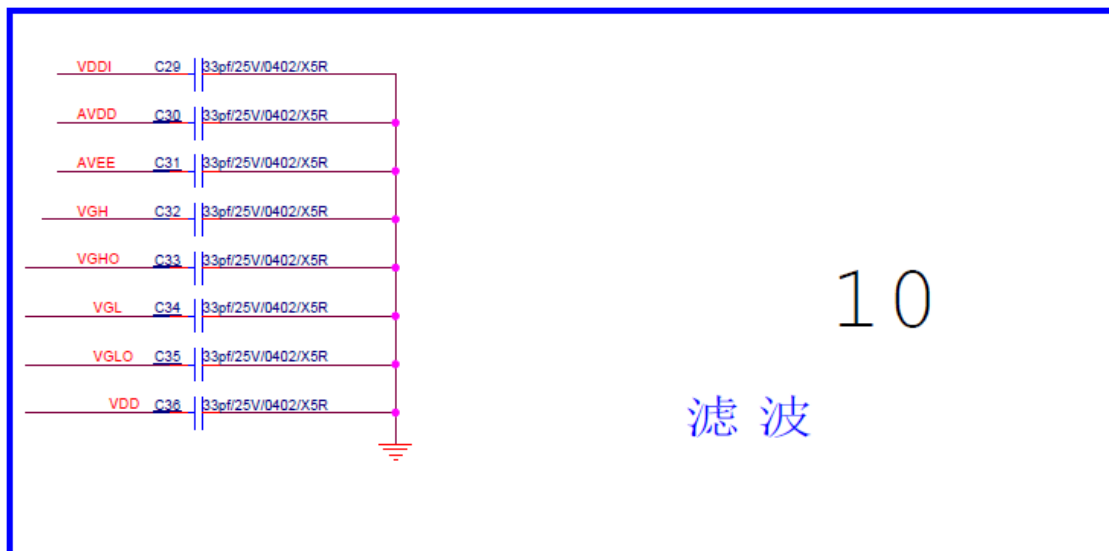
CAPACITOR

Layout near Master IC



Layout near Slaver IC

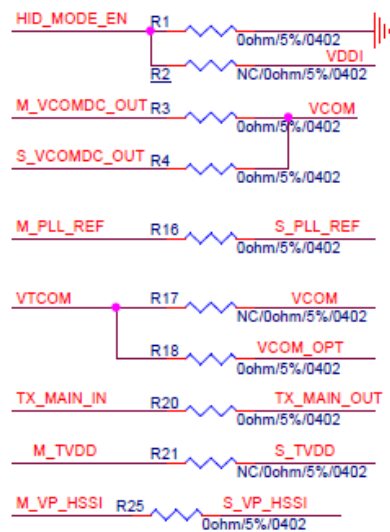






Resistor

Resistor



Diode

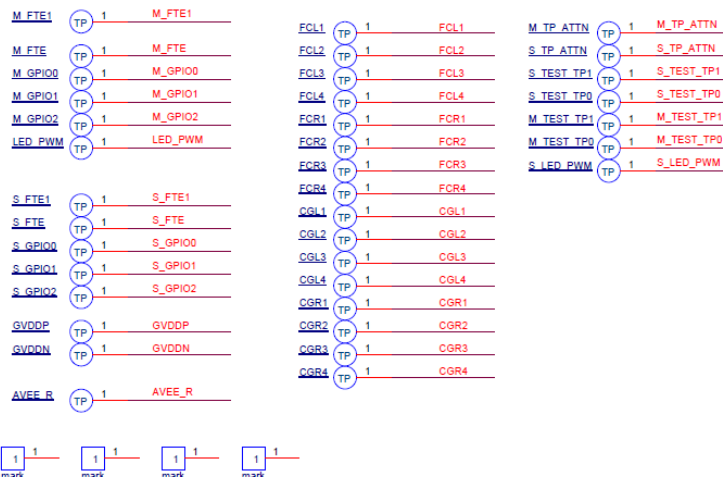
ESD



Test Point

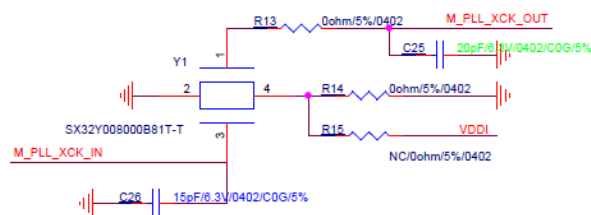


TEST PADS



Pen

W/I PEN



Reserved for MIPI Protect

TBD



3. FOG Pin Assignment

Compatible Driver IC: [NT36532](#)

Note 1: Must be used according to related IC spec;

J1: [20982-051E-01](#)

Pin No.	Symbol	Description	I/O
1	GND	Ground	P
2	GND	Ground	P
3	NC	NC	-
4	AVDD	Negative input analog power for driver IC	P
5	AVDD	Negative input analog power for driver IC	P
6	AVDD	Negative input analog power for driver IC	P
7	NC	NC	-
8	AVEE	Positive input analog power for driver IC	P
9	AVEE	Positive input analog power for driver IC	P
10	AVEE	Positive input analog power for driver IC	P
11	NC	NC	-
12	GND	Ground	P
13	GND	Ground	P
14	D3-	MIPI differential data3 input (Negative)	I
15	D3+	MIPI differential data3 input (Positive)	I
16	GND	Ground	P
17	D0-	MIPI differential data0 input (Negative)	I
18	D0+	MIPI differential data0 input (Positive)	I
19	GND	Ground	P
20	CLK-	MIPI differential clock input (Negative)	I
21	CLK+	MIPI differential clock input (Positive)	I
22	GND	Ground	P
23	D1-	MIPI differential data1 input (Negative)	I
24	D1+	MIPI differential data1 input (Positive)	I
25	GND	Ground	P
26	D2-	MIPI differential data2 input (Negative)	I
27	D2+	MIPI differential data2 input (Positive)	I
28	GND	Ground	P

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29	GND	Ground	P
30	ID0	System Hardware ID0 Select(Pull Low to VDDI with 10KΩ)	O
31	ID1	System Hardware ID1 Select(Pull Low to GND with 10KΩ)	O
32	LEDPWM_OUT	Output pin for PWM signal of LED driving.	O
33	TE	TE	I
34	REST	This signal will reset the device and must be applied properly to initialize the chip.	I
35	GND	Ground	P
36	GND	Ground	P
37	NC	NC	-
38	VDD18	Power Supply 1.8V	P
39	VDD18	Power Supply 1.8V	P
40	VDD18	Power Supply 1.8V	P
41	NC	NC	-
42	GND	Ground	P
43	TP_INT	Interrupt signal for TP	I
44	TP_RST	The external reset input for TP	I
45	TP_SPI_CS	TP:SPI CS(TYP. 1.8V)-for TP	I
46	TP_SPI_SCLK	TP:SPI SCLK(TYP. 1.8V)-for TP	I
47	TP_SPI_MISO	TP:SPI MISO(TYP. 1.8V)-for TP	I
48	TP_SPI_MOSI	TP:SPI MOSI(TYP. 1.8V)-for TP	I
49	GND	Ground	P
50	GND	Ground	P
51	GND	Ground	P

J2: 20655-011E-01

Pin No.	Symbol	Description	I/O
1	FB1	Power for LED1 Cathode	P
2	FB2	Power for LED2 Cathode	P
3	FB3	Power for LED3 Cathode	P
4	FB4	Power for LED4 Cathode	P
5	FB5	Power for LED5 Cathode	P
6	FB6	Power for LED6 Cathode	P



7	FB7	Power for LED7 Cathode	P
8	FB8	Power for LED8 Cathode	P
9	NC	NC	-
10	VLED	Power for LED Anode	P
11	VLED	Power for LED Anode	P

J3: 20599-013E-01

Pin No.	Symbol	Description	I/O
1	LEDA1	Power for LED Anode	P
2	NC	NC	-
3	LEDA2	Power for LED Anode	P
4	NC	NC	-
5	NC	NC	-
6	LEDK1	Power for LED1 Cathode	P
7	LEDK3	Power for LED3 Cathode	P
8	LEDK5	Power for LED5 Cathode	P
9	LEDK7	Power for LED7 Cathode	P
10	LEDK2	Power for LED2 Cathode	P
11	LEDK4	Power for LED4 Cathode	P
9	LEDK6	Power for LED6 Cathode	P
10	LEDK8	Power for LED8 Cathode	P



4 Electrical Specification

4.1 Absolute maximum ratings

LEDK=GND=0V, Ta = 25℃

Item	Symbol	Min	Max	Unit	Remark
Power Voltage	IOVCC	1.65	1.95	V	
	LED+	2.7	3.0	V	
Operating Temperature	TOPR	-20	60	℃	
Storage Temperature	TSTG	-40	80	℃	

4.2 Recommended Operating Condition

AVDD=5.5V,AVEE=-5.5V,IOVCC=1.8V,GND=0V, Ta =

25℃

Item	Symbol	Min.	Typ.	Max.	Unit	Remark
Logic Supply Voltage	VIO	1.65	1.8	1.95	V	Refer to IC ILB
Analog Supply Voltage	VSP	5.7	5.8	5.9	V	
	VSN	-5.7	-5.8	-5.9	V	

4.3 Current of 3 power

AVDD=5.5V,AVEE=-5.5V,IOVCC=1.8V,GND=0V, Ta =

25℃

Item	Symbol	Min	Typ	Max	Unit	Remark
Current of 3 Power @White 60Hz with touch	IOVCC	-	78	130	mA	
	VSP	-	37	68	mA	
	VSN	-	18	38	mA	

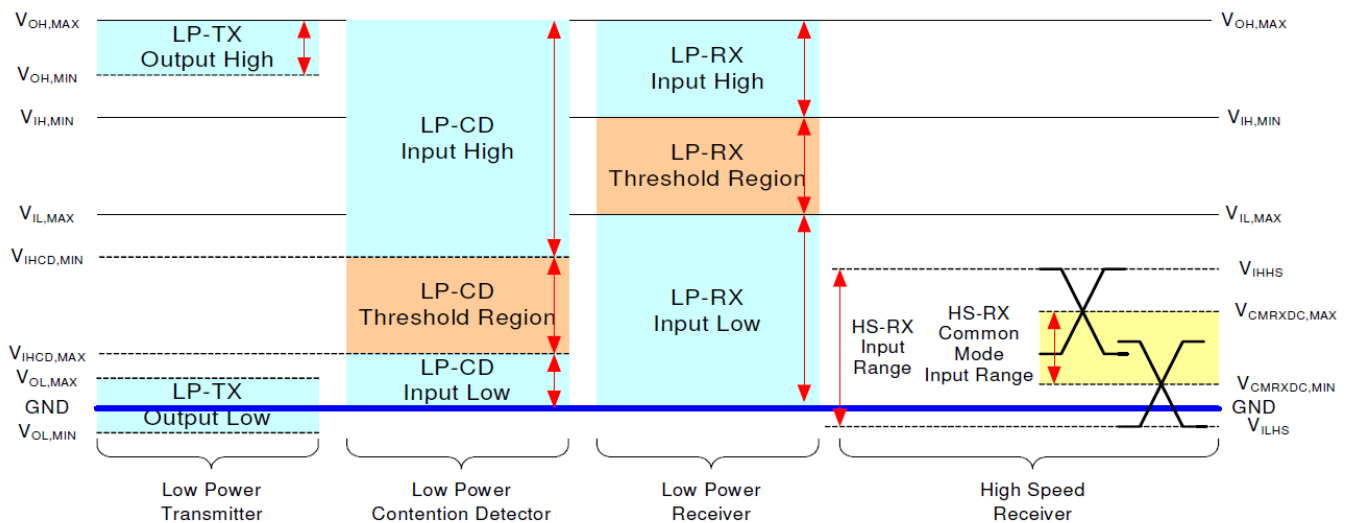
4.3MIPI Interface Characteristics

(a) MIPI interface DC characteristic:

Item		Parameter	Min	Typ	Max	Unit
LP _T W	Thevenin output high level	VOH	1.1	1.2	1.3	V
	Thevenin output low level	VOL	-50		50	mV
	Output impedance of LP transmitter	ZOLP	45	50	55	Ω
HS	Common-mode voltage HS receive mode	VCMRX(DC)	70	-	330	mV

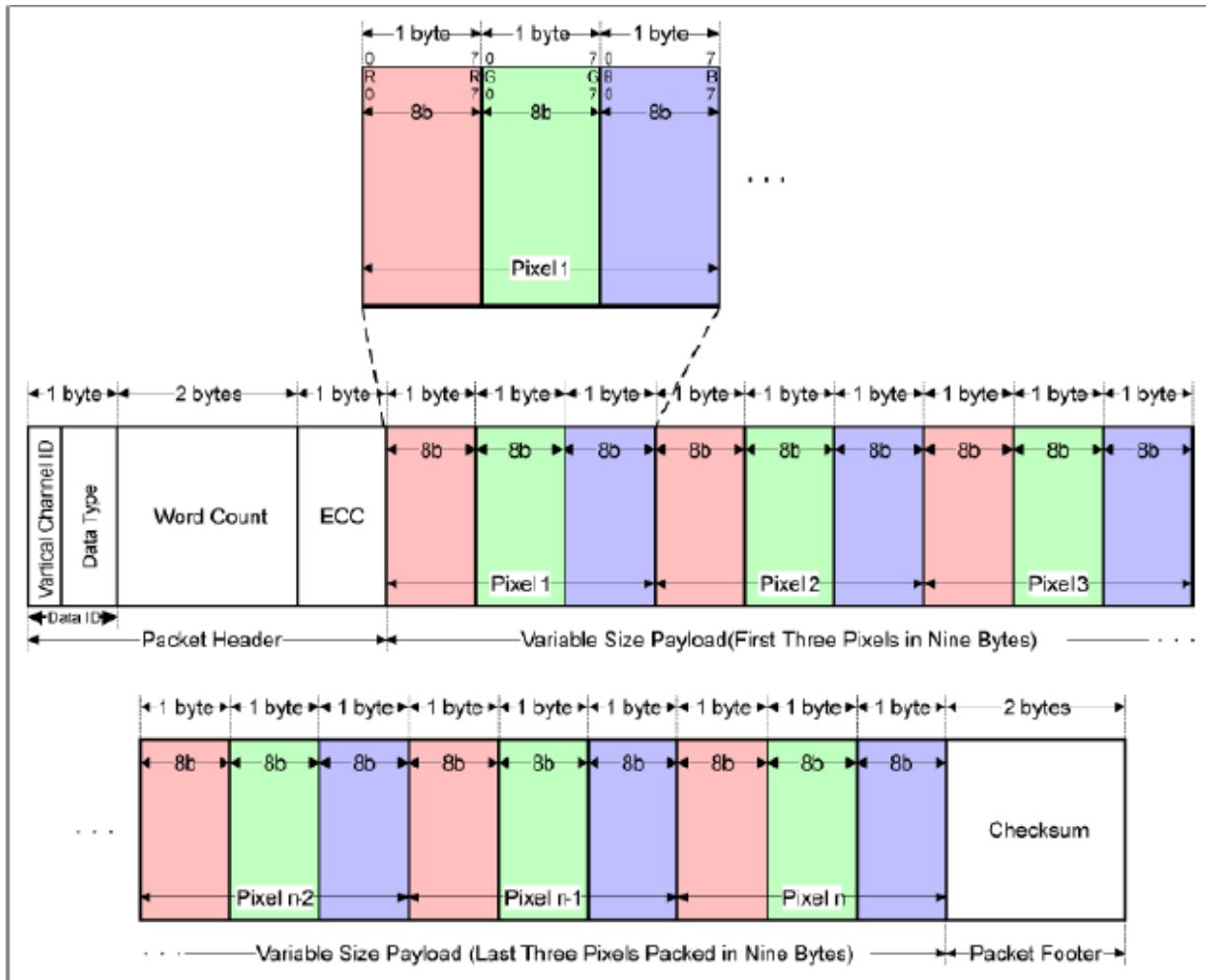


R X	Differential input high threshold	VIDTH	-	-	70	mV
	Differential input low threshold	VIDTL	-70	-	-	mV
	Single-ended input high voltage	VIHHS	-	-	460	mV
	Single-ended input low voltage	VILHS	-40	-	-	mV
	Single-ended threshold for HS termination enable	VTERM-EN	-	-	450	mV
	Differential input impedance	ZID	80	100	125	Ω
LP R X	Logic 1 input voltage	VIH	880	-	I0VCC	mV
	Logic 0 input voltage. not in ULPState	VIL	0	-	550	mV
	Input hysteresis	VHYST	25	-	-	mV
LP C D	Logic 1 contention threshold	VIHCD	450	-	I0VCC	mV
	Logic 1 contention threshold	VILCD	0	-	200	mV





4.4 MIPI 24 bits RGB Data Format



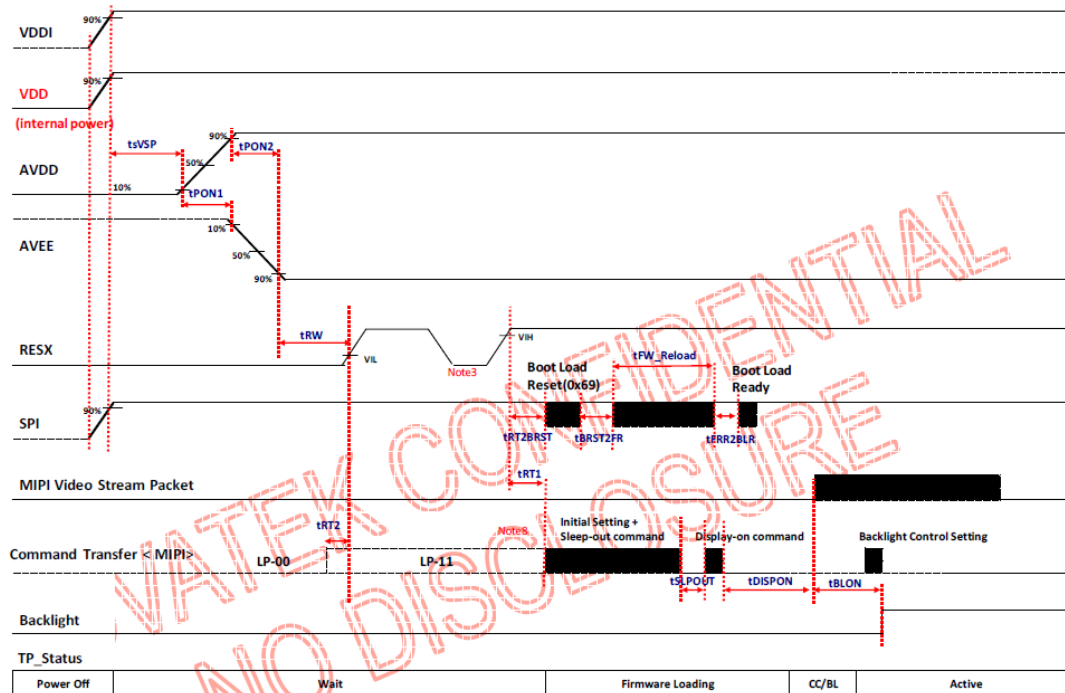


4.5 Power on/off sequence

Power on :

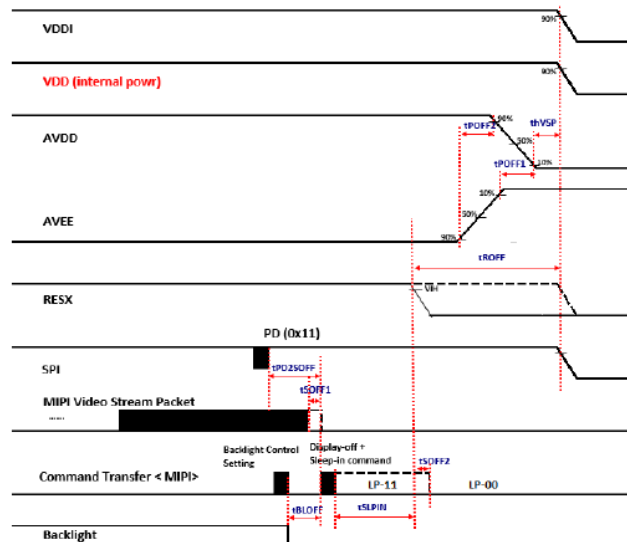
5.9.1.1 Three power (VDDI/ AVDD/ AVEE) – No Flash Mode

<Power on>



Power off :

<Power off>



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Item	Symbol	Condition	Min	Typ	Max	Unit
AVDD - AVEE delay time (10% to 10%)	tPON1	Power On	0	-	-	us
AVDD - AVEE delay time (90% to 90%)	tPON2	Power On	0	-	-	us
VDDI On to AVDD On time (90% to 10%)	tsVSP	Power On	1	-	-	ms
All Power On to RESX Hi-level time (90% to VIL)	tRW	Power On	10	-	-	ms
RST Hi-level (VIH) to SPI Boot reset time	tRT2BRST	Power On	10	-	-	ms
Boot load reset command to FW reload	tBRST2FR	Power On	5	-	-	ms
FW reload	tFW_Reload	Power On	65	-	-	ms
FW reload to Boot load ready time	tFR2BLR	Power On	0	-	-	ms
RST Hi-level (VIH) to 1st Command time	tRT1	Power On	10	-	-	ms
(MIPI goes to LP11 state) to RST goes to Low-level (VIL)	tRT2	Power On	-	0	-	ms
SLPOUT Sequence Request time	tSLPOUT	Power On	100	-	-	ms
MIPI Video Stream On time to BL On time	tBLON	Power On	40	-	-	ms
AVDD - AVEE delay time (10% to 10%)	tPOFF1	Power Off	0	-	-	us
AVDD - AVEE delay time (90% to 90%)	tPOFF2	Power Off	0	-	-	us
AVDD Off to VDDI Off time (10% to 90%)	tsVSP	Power Off	0	-	-	us
RST Low to VDDI Off time (VIH to 90%)	tROFF	Power Off	0	-	-	us
BL Off to Display Off Command time	tBLOFF	Power Off	0	-	-	us
SLPIN Sequence Request time	tSLPIN	Power Off	60	-	-	ms
VDD regulator turn off time	tVDDOFF	Suspend	80	-	-	ms
MIPI Video Stream Off time to DISPLAY OFF command	tSOFF1	Power Off	0	-	-	us
SPI Power Down to DISPLAY OFF command	tPD2SOFF	Power Off	50	-	-	ms
SPI Wake Up Gesture to DISPLAY OFF command	tWKG2SOFF	Power Off	50	-	-	ms



5 Optical Characteristics

a) Optical Specification

Ta=25℃

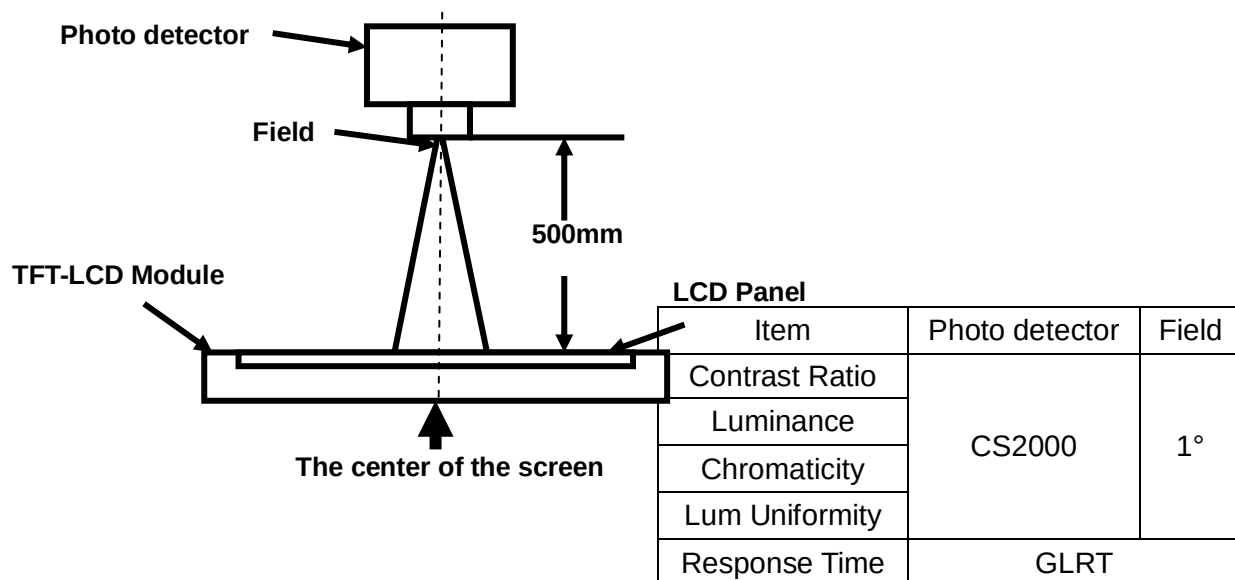
Item		Symbol	Condition	Min.	Typ.	Max.	Unit	Device	Remark
Luminance		Bp	$\theta=0^{\circ}$ $\Phi=0^{\circ}$	350	400	-	cd/m2	CS2000	Center point value
Luminance Uniformity (13P)		Δ Bp	$\theta=0^{\circ}$ $\Phi=0^{\circ}$	75	80	-	%	CS2000	
σ Lv (13P)		-	$\theta=0^{\circ}$ $\Phi=0^{\circ}$	-	-	30	%	CS2000	Lv:((Max(13p)-Min(13p))/Avg(13p))
NTSC DCI-P3 Coverage (1976)		N	$\theta=0^{\circ}$ $\Phi=0^{\circ}$	91	96	-	%	CS2000 /SR3	Note 2
Contrast Ratio		CR	$\theta=0^{\circ}$	1000	1500	-	NA	CS2000 /SR3	Note1/3
View Angles		θ T	$CR \geq 10$	80	85	-	Degree	EZcontrastXL88	Note2
		θ B		80	85	-			
		θ L		80	85	-			
		θ R		80	85	-			
CCT Range (13P)		-	$\theta=0^{\circ}$ $\Phi=0^{\circ}$	-	-	670	K	CS2000	
Gamma		-	-	1.9	2.2	2.5			
White Color shift		-	θ T=30	-	-	3	JNCD		
		-	θ B=30	-	-	3	JNCD		
		-	θ L=30	-	-	3	JNCD		
		-	θ R=30	-	-	3	JNCD		
Flicker		-	50Hz @127Gray	-	-	-30	dB		
Crosstalk		Δ CT	-	-	-	2	%		
Response Time		Ton+Toff	25℃	-	20	25	ms	GLRT	Note4
Chromaticity	White	x	$\theta=0^{\circ}$ Ta=25℃	0.265	0.295	0.325		CS2000	Note5
		y		0.285	0.315	0.345			
	Red	x		0.644	0.674	0.704			
		y		0.291	0.321	0.351			
	Green	x		0.240	0.270	0.300			
		y		0.652	0.682	0.712			
	Blue	x		0.111	0.141	0.171			
		y		0.034	0.064	0.094			

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Note 1: Definition of optical measurement system.

The optical characteristics should be measured in dark room. After 5 minutes operation, the optical properties are measured at the center point of the LCD screen. All input terminals LCD panel must be ground when measuring the center area of the panel.



Note 2: Definition of viewing angle range and measurement system.

viewing angle is measured at the center point of the LCD by EZcontrastXL88.

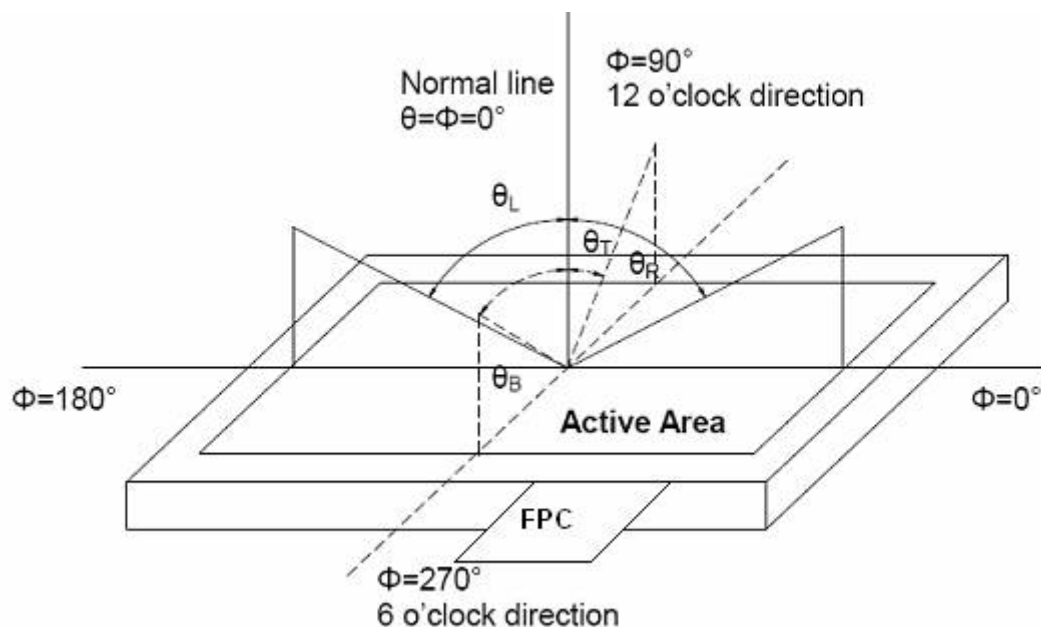


Fig. 1 Definition of viewing angle



Note 3: Definition of contrast ratio

$$\text{Contrast ratio (CR)} = \frac{\text{Luminance measured when LCD is on the "White" state}}{\text{Luminance measured when LCD is on the "Black" state}}$$

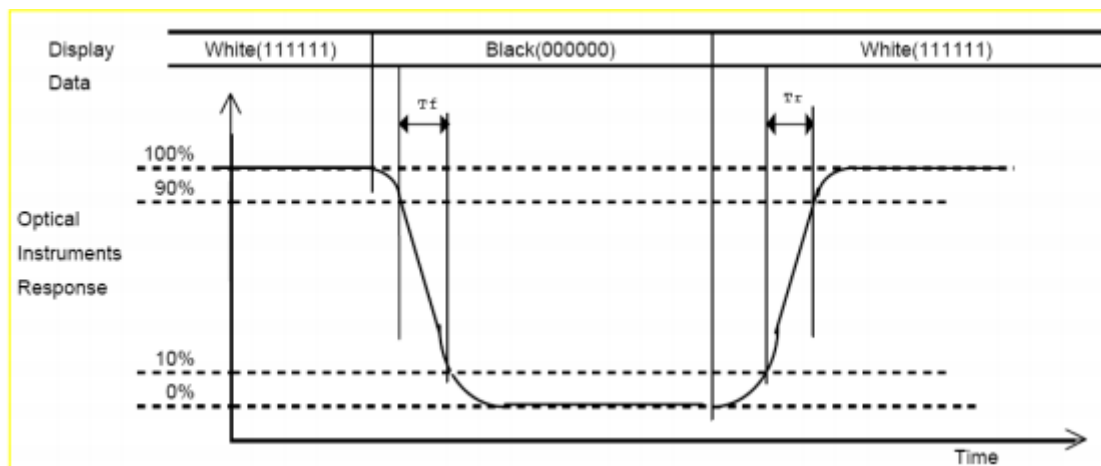
“White state”: The state is that the LCD should be driven by V_{white}.

“Black state”: The state is that the LCD should be driven by V_{black}.

V_{white}: To be determined V_{black}: To be determined.

Note 4: Definition of Response time

The response time is defined as the LCD optical switching time interval between “White” state and “Black” state. Rise time (T_r) is the time between photo detector output intensity changed from 10% to 90%. And fall time (T_f) is the time between photo detector output intensity changed from 90% to 10%.



Note 5: Definition of color chromaticity (CIE1931)

Color coordinates measured at center point of LCD

“Black state”: The state is that the LCD should driven by V_{black}.

V_{white}: To be determined V_{black}: To be determined.



6 Environmental / Reliability Test

测定项目		规格	
光学特性	1	Brightness	Min.350,Typ.400
		White balance	(0.295,0.315)+/-0.03
		Contrast Ratio	Min.1000,Typ.1500
		Color gamut	Min91%, Typ. 96%
	2	Brightness uniformity	13p>80%
		White balance uniformity	0.0100
	3	Viewing angle	Min 75° ,Typ80°
	4	Color shift	Δ ≤0.012
	5	Gamma	1.9~2.5
	6	Flicker	' -30dB
	7	Crosstalk	Max 2%
	8	Response time	Typ. 20, Max. 25
	9	Image sticking	1min 内消失
10	待机闪屏测试	无闪屏	
环境 信赖性实验	11	高温驱动（HTO）	无不良
	12	低温驱动（LTO）	无不良
	13	高温高湿驱动（HTBO）	无不良
	14	高温存储（HTS）	无不良
	15	低温存储（LTS）	无不良
	16	高温高湿存储（HTBS）	无不良
	17	寿命测试（ALT）	无不良
	18	Thermal shock test	无不良
	19	Cold bubble	无 bubble
	20	Altitude test	无不良
结构强度 信赖性实验	21	LCM 4 轴 bending	ε 置信下限≥1500 μ
	22	Panel 4 轴 bending	ε 置信下限≥1500 μ
	23	Ring On Ring（参考项）	至样品破裂
	24	面压测试	显示正常
	25	Cell gap 强度	显示正常
	26	黄斑测试	显示正常
	27	vacuum test	显示正常
	28	FOG 拉力强度	≥7N/cm
	29	FOG 柔韧度测试	δ ≤10%
	30	IC 3 轴 bending	B10 ≥ 350Mpa
	31	小球跌落试验	显示正常
电性测试	32	Bonding resistance	R _{CFOG} ≤ 27 Ω
			R _{FOG} ≤ 12 Ω
			R _{COG} ≤ 15 Ω
	33	功耗测试	
	34	ESD test	测试后显示无异常
	35	EMC	
36	Inrush current test	Follow 低电行为规范	

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7 Packing Method

Module Face Down and
Then Put 2 LCMs into a Tray



Stacked 9 +1 empty tray
Between the Trays Should
Rotate 180 Degrees



Use tape
binding the
trays Like the
right fig



Use Desiccant *2 Per Bag



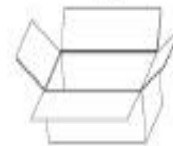
Put trays & Desiccants into
an Anti statistic bag



Seal the carton And
Stick a Tianma Label;



Put 2 boxes
into a carton



Stick a Label



Put the bag
into a box



Folding the
bag



(1) 纸箱堆叠数按 2*3/每层*共 5 层

